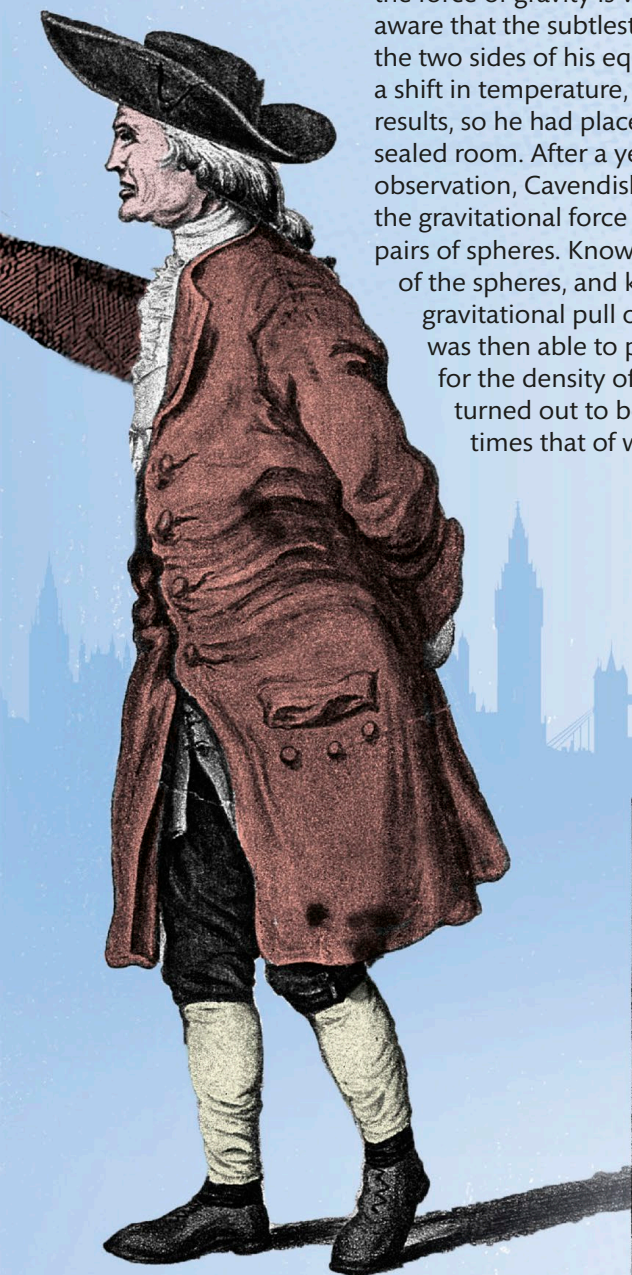




rod. Cavendish viewed the oscillations on the rod through a telescope; because the force of gravity is very weak, he was aware that the subtlest disturbance in the two sides of his equipment, even a shift in temperature, might skew the results, so he had placed it in a dark, sealed room. After a year of painstaking observation, Cavendish determined the gravitational force between the pairs of spheres. Knowing the density of the spheres, and knowing the gravitational pull of the Earth, he was then able to produce a value for the density of the Earth, which turned out to be more than five times that of water.

**CONDUCTED
MORE THAN 400
INVESTIGATIONS
INTO THE NATURE OF AIR**

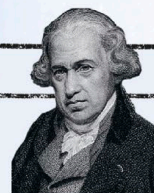
**PROVED THAT
THE DENSITY
OF EARTH WAS
5 TIMES THAT
OF WATER**



**\$8,230 OF HIS LEGACY
WAS USED TO BUILD THE
CAVENDISH LABORATORY
AT CAMBRIDGE**

JAMES WATT

British inventor, engineer, and chemist James Watt's work on the composition of water led to a long-running "water controversy."



In 1783, Watt (1736–1819) and Cavendish reached almost the same conclusion about the chemical make-up of water. But whether Watt, Cavendish, or French chemist Antoine Lavoisier (see pp.100–103) deserved the most credit for this discovery became the subject of debate. Cavendish performed his experiments first but published them after Watt's paper. Some historians say that Watt identified water as a chemical compound more explicitly than Cavendish. However, Lavoisier was the first to state clearly that water is composed of hydrogen and oxygen rather than "phlogiston" (the term used by Watt and Cavendish). Watt is best known for his patented improvements to steam engines.